

Distributed RAG systems process documents asynchronously.

Redis queues background jobs.

FastAPI handles upload APIs.

Vector databases store semantic embeddings for efficient retrieval.

Workers consume queued tasks independently for scalability.

FAISS enables fast similarity search across embedded documents.

Chunking pipelines split large documents into manageable sections.

Docker containers isolate services for consistent deployment environments.

Microservice architecture improves maintainability and fault isolation.

Embedding models transform raw text into machine-understandable vectors.

Background ingestion pipelines preprocess and index uploaded documents.

RESTful APIs enable communication between distributed backend services.